
Tripuraneni Gayathri Praharshita

Data Analyst



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OBJECTIVE

Detail-oriented Data Analyst with 5+ years of experience in leveraging data to uncover trends, optimize performance, and support key business decisions. Skilled in data mining, dashboard development, and predictive modelling using tools like SQL, Python, Tableau, and Power BI. Seeking to contribute analytical expertise and strategic insight to a dynamic organization focused on innovation and data-driven excellence.

PROFILE SUMMARY

- Data Analyst with **5+** years of experience in analysing complex datasets to uncover business insights, proficient in **SQL, Python, R, and Excel**, with strong data extraction, transformation, and cleaning capabilities.
- Skilled in designing and developing interactive dashboards using **Tableau, Power BI, and Google Data Studio**, enabling stakeholders to make real-time, data-driven decisions.
- Expertise in statistical analysis, predictive modelling, and **machine learning** using tools like **Scikit-learn, TensorFlow, and XGBoost** to identify trends and forecast outcomes accurately.
- Adept at working with relational databases such as **MySQL, PostgreSQL, and Oracle**, and experienced in handling **NoSQL databases** like **MongoDB** for unstructured data solutions and hands-on experience with **ETL tools** like **Informatica, Talend, SSIS, Pentaho, and Apache NiFi** for building and maintaining robust data pipelines across multiple systems and strong knowledge of data warehousing tools including **Snowflake, Amazon Redshift, and Google Big Query**, optimizing data storage and retrieval for analytics workflows.
- Experienced in implementing cloud-based solutions using **AWS S3, Redshift, Athena, Azure Data Lake, and Azure Synapse Analytics** to build scalable data environments.
- Proficient in using API integration techniques with tools like **Postman** and libraries such as **Requests** and **Beautiful Soup** to retrieve data from various web and application sources and solid command over scripting with **Bash** and **Shell**, automating routine data processes and enhancing workflow efficiency within distributed systems
- Proven ability to implement **CI/CD pipelines** using **Jenkins, GitLab CI/CD**, and containerized applications through **Docker** for seamless and continuous deployment and extensive experience in building real-time data pipelines and stream processing using **Apache Kafka, Apache Spark, and Apache Flink**, ensuring fast and reliable data delivery.
- Strong in data processing and visualization using **Pandas, NumPy, Matplotlib, Seaborn, and Plotly**, enabling detailed analysis and effective presentation of business metrics.
- Expertise in using the **ELK Stack (Elasticsearch, Logstash, Kibana)** to collect, index, and visualize large-scale data logs and developed dashboards in Kibana to monitor system performance and detect anomalies.
- Knowledgeable in workflow orchestration tools like **DBT, Apache Airflow** and **Luigi** for pipeline management and designed and automated complex ETL workflows supporting both batch and streaming data.
- Familiar with **DevOps practices** such as CI/CD integration, containerization, and environment versioning and Managed codebase using **Git/GitHub** and maintained collaboration via pull requests and branching strategies.
- Proficient in **SQL** and **NoSQL** databases including **PostgreSQL, MySQL, MongoDB, and Cassandra** for complex query design and developed stored procedures and optimized queries to improve performance in analytical workloads.
- Skilled in using **Python** libraries such as **Pandas, NumPy, and Scikit-learn** for data analysis and machine learning and built predictive models and automated data transformation pipelines to support decision-making.

TECHNICAL SKILLS

- **Data Analysis & Visualization:** Tableau, Power BI, Looker, Google Data Studio, Matplotlib, Seaborn, Plotly, D3.js, Chart.js, Excel (Advanced), Google Sheets
- **Programming Languages:** **Python** (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Plotly) **R** (ggplot2, dplyr, caret) **SQL** (MySQL, PostgreSQL, SQL Server), Oracle, SQLite, **JavaScript**
- **Database Management:** MySQL, PostgreSQL, SQL Server, Oracle, **NoSQL databases** (MongoDB, Cassandra)
- **Data Warehousing:** Snowflake, Amazon Redshift, Google Big Query
- **ETL Tools:** Informatica, Talend, SSIS, Apache NiFi, Pentaho, SAP, SAP HANA

- **Big Data & Cloud Technologies:** Hadoop, Apache Spark, Kafka, **AWS** (S3, Redshift, Athena, Lambda), **Azure** (Azure Data Factory, Synapse Analytics, Databricks, Blob storage), **Google Cloud Platform (GCP)** (Big Query, Dataflow, Pub/Sub, Cloud Storage, Bash)
- **Machine Learning & Statistical Analysis:** Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost
- **Statistical Analysis:** Hypothesis testing, A/B testing, Regression analysis, cohort analysis, Time Series Forecasting, Exploratory Data Analysis(EDA), Data Profiling, Statsmodels
- **Data Engineering & Automation:** Apache Airflow, Jenkins, Docker, Kubernetes (CI/CD), DataOps, Git, GitHub, DBT
- **Data Integration & APIs:** RESTful APIs, GraphQL, Postman, Swagger
- **Web Scraping:** BeautifulSoup, Scrapy, Selenium
- **Data Validation:** Great Expectations, Deequ
- **Other Tools & Technologies:** Jupyter Notebooks, Google Colab, Elasticsearch, Kibana (ELK Stack), JDBC/ODBC

EDUCATION

Lamar University, Masters in Management Information Systems, USA from 2023 - 2024

WORK EXPERIENCE

Trinity Industries, Inc, Dallas, Texas, USA

Senior Data Analyst | Jul 2024 – Present

Description: Trinity Industries, Inc. is a leading provider of rail transportation products and services. Lead the analysis of complex data sets to generate actionable insights that support key business decisions and lead and manage data analysis projects, ensuring timely delivery of reports and recommendations in alignment and maintaining and managing databases to ensure data accuracy, accessibility, and integrity.

Responsibilities:

- Analysed complex rail transportation and logistics data using **SQL** and **Python** to identify inefficiencies in fleet utilization and support data-driven operational improvements across manufacturing and leasing operations.
- Developed and maintained scalable **ETL pipelines** to extract, transform, and load data into **SQL Server** and **Azure Synapse Analytics**, enabling accurate reporting on railcar maintenance, leasing cycles, and asset tracking.
- Designed interactive dashboards using **Power BI** and **Tableau** to visualize key metrics such as turnaround time, asset availability, repair frequency, and fleet allocation, improving visibility for stakeholders.
- Leveraged **machine learning** models using **scikit-learn**, **TensorFlow**, and **R** to predict railcar failure risks, optimize maintenance schedules, and forecast leasing demand trends across industry sectors.
- Processed large-scale telemetry and sensor data from railcars using **Hadoop** and **Apache Spark**, enabling advanced analytics on usage patterns, track conditions, and load performance.
- Conducted statistical analysis and data mining using **Python** and **R**, applying clustering, regression, and classification to optimize route planning, fuel usage, and operational efficiency, applied **NLP** techniques using **spaCy** and **NLTK** to analyse unstructured maintenance logs and technician notes, extracting key failure modes and improving diagnostic accuracy.
- Monitored and maintained data pipeline health using **Apache Airflow**, ensuring consistent and timely delivery of railcar performance and leasing data across platforms and designed data models in **Snowflake** to support scalable, centralized analytics for fleet operations, maintenance history, and customer leasing contracts.
- Utilized **MongoDB** and **Cassandra** for managing unstructured datasets such as inspection reports, image metadata, and shipment documents, enabling faster retrieval and reporting and created advanced data visualizations using **Looker** to present insights into railcar lifecycle management, maintenance ROI, and route optimization for executive reporting.
- Ensured data accuracy and quality through profiling and validation using **Talend** and **Alteryx**, supporting regulatory compliance and operational standards and integrated data across systems using **Informatica** to support seamless synchronization between on-premise ERP systems, cloud-based platforms, and field data sources.
- Containerized machine learning and analytics applications using **Docker** and deployed them using **Kubernetes** to support scalable infrastructure across rail operations and implemented **CI/CD pipelines** using **Jenkins**, **GitLab**, and **Bitbucket** to automate deployment of analytics solutions and monitoring tools across Trinity's digital platforms.

Environment: SQL, Python, Azure Synapse Analytics, Power BI, Tableau, scikit-learn, TensorFlow, Hadoop, Apache Spark, Apache Kafka, Azure Data Factory, spaCy, NLTK, Apache Airflow, Snowflake, MongoDB, Cassandra, Looker, Talend, Alteryx,

Informatica, Docker, Kubernetes, CI/CD pipelines, Jenkins, GitLab, Bitbucket, Jenkins, Git, Power BI, Tableau, Looker, GeoPandas, QGIS, PostGIS, XGBoost, LightGBM, Cat Boost, OpenCV, YOLOv5, Apache NiFi, Flume, Kafka Streams.

Tesla, Inc., Austin, Texas, USA

Senior Data Analyst | Oct 2023 – Jun 2024

Description: Tesla, Inc. is a global leader in electric vehicles, clean energy, and sustainable transportation solutions company designs and manufactures electric cars, battery energy storage, and solar products. Managing and organizing data within databases and systems and developing and maintaining data models and dashboards and creating visual representations of data (charts, graphs, dashboards) to communicate findings effectively.

Responsibilities:

- Collected, extracted, and integrated operational, sensor, and performance data from vehicle systems, manufacturing units, and energy platforms using **SQL, Python, and APIs** for end-to-end analytics across production and energy operations.
- Performed data cleansing, transformation, and validation using **Python (Pandas, NumPy), R, and SQL** to ensure high-quality datasets supporting predictive maintenance, energy efficiency, and vehicle telemetry analysis.
- Designed and developed dashboards and performance reports using **Tableau, Power BI, and Excel** to monitor key metrics such as battery health, fleet performance, and production throughput.
- Managed cloud data storage across platforms such as **Amazon Redshift, S3, and Google Big Query**, optimizing data retrieval from vehicle logs, solar performance systems, and manufacturing IoT streams.
- Built and automated scalable ETL pipelines using **Apache Airflow, Talend, and AWS Glue**, streamlining integration of vehicle diagnostics, renewable energy output, and real-time production data.
- Developed real-time data processing pipelines using **Apache Kafka and Spark**, enabling instant analysis of vehicle telemetry, driver behaviour, and manufacturing line anomalies.
- Applied machine learning techniques with **Scikit-learn, TensorFlow, and Keras** to forecast energy consumption, detect anomalies in vehicle performance, and optimize supply chain logistics.
- Conducted financial modelling and cost optimization using **Excel and Python** to support energy project budgeting, long-term forecasting, and electric vehicle profitability analysis.
- Automated regulatory compliance and performance reporting using **Python and BI tools**, ensuring adherence to safety, sustainability, and operational standards across vehicle and energy platforms.
- Integrated diverse data sources including vehicle systems, battery diagnostics, solar generation data, and CRM records using **Informatica, SSIS, and Talend** for unified reporting and analysis.
- Leveraged **Hadoop** for large-scale storage and distributed processing of telemetry, sensor, and energy usage data to support engineering, analytics, and research teams and deployed machine learning models using **Scikit-learn, TensorFlow, and Python** to support use cases like autonomous driving insights, customer segmentation.
- Designed and maintained high-performance data warehousing solutions using **Snowflake**, enabling efficient querying and data modelling across large datasets from Tesla's energy and automotive divisions.
- Created interactive dashboards using **Looker** for real-time tracking of production metrics, charging network status, and solar panel performance, supporting executive-level decision-making.

Environment: SQL, Python, APIs, Pandas, NumPy, R, Tableau, Power BI, Excel, Amazon Redshift, **S3**, Google Big Query, Apache Airflow, Talend, AWS Glue, Scikit-learn, TensorFlow, Keras, Informatica, Apache Spark, SSIS, Talend, Hadoop, Snowflake, Matplotlib, Looker

Dr Reddy's Laboratories Ltd. (Capgemini), Hyderabad, India

Data Analyst | Aug 2021 – Jun 2023

Description: Dr Reddy's Laboratories Ltd. is a leading global pharmaceutical company specializing in the development, manufacturing, and marketing of a wide range of generic medications, active pharmaceutical ingredients. Developing and maintaining databases and data systems and Ensuring data accuracy, completeness, and consistency through data cleaning and validation processes.

Responsibilities:

- Designed and implemented robust **data models** and automated **ETL pipelines** using **SQL Server, Snowflake, and Python** to manage R&D, production, and quality control datasets, ensuring accuracy, consistency, and availability of critical pharmaceutical data.

- Developed interactive **dashboards** and **analytics reports** using **Power BI**, **Tableau**, and **Looker** to track manufacturing KPIs, batch yields, and supply chain efficiency, enabling data-driven decision-making for production
- Built and optimized complex **SQL queries**, **stored procedures**, and data transformation routines to aggregate large-scale experimental and production data, supporting compliance reporting and process validation.
- Automated data integration from **LIMS**, **ERP**, **CRM**, and third-party systems using **Informatica**, **Talend**, **Alteryx**, and **Microsoft SSIS**, enabling seamless consolidation of structured and semi-structured pharmaceutical and clinical datasets.
- Developed predictive **machine learning models** with **scikit-learn**, **TensorFlow**, and **Keras** to forecast drug stability, optimize production schedules, detect process anomalies, and improve formulation success rates.
- Leveraged **Apache Kafka** and **Apache Spark** to build real-time streaming pipelines for monitoring production lines, lab experiments, and supply chain alerts, ensuring timely detection of deviations and regulatory compliance.
- Processed and analyzed large-scale pharmaceutical data using **Hadoop**, **Hive**, and **Spark** to identify trends in drug efficacy, inventory movement, and batch quality, supporting operational efficiency and continuous improvement initiatives.
- Implemented **CI/CD pipelines** using **Jenkins**, version control with **Git**, and automated deployment of analytics and machine learning solutions, ensuring reliable updates to R&D, production, and regulatory reporting systems.
- Integrated cloud platforms such as **AWS S3**, **AWS Redshift**, and **Google Big Query** to enable secure storage, high-performance analytics, and scalable access to research, production, and regulatory datasets for cross-functional teams.
- Designed and maintained advanced **data processing workflows** in **Snowflake**, facilitating efficient reporting on batch metrics, process deviations, regulatory compliance, and quality control across global pharmaceutical operations.

Environment: ETL pipelines, SQL Server, Snowflake, LIMS, ERP, CRM, Informatica, Talend, Alteryx, scikit-learn, TensorFlow, Keras, Apache Kafka, Apache Spark, Hadoop, Hive, Spark, CI/CD pipelines, Jenkins, Git, AWS S3, AWS Redshift, Google Big Query, Snowflake.

Axis Max Life Insurance, Hyderabad, India

Data Analyst | Aug 2019 – July 2021

Description: Axis Max Life Insurance is one of the leading life insurance companies offering a wide range of life insurance solutions to individuals and families. Responsible for collecting and analysing data to provide valuable insights and manage databases, troubleshoot issues, and optimize performance and generate reports and present findings to aid decision-making and maintaining and managing databases.

Responsibilities:

- Created interactive **dashboards** and **reports** using **Tableau**, **Power BI**, and **Excel** to monitor policy issuance, claims processing efficiency, and customer service KPIs, enabling actionable insights for management across multiple business
- Built and tested predictive **models** using **Python**, **R**, **scikit-learn**, and **TensorFlow** to forecast policy lapse rates, optimize premium collection strategies, and enhance customer retention across life insurance products.
- Wrote complex **SQL queries** in **MySQL** and **SQL Server** to extract, aggregate, and analyse policyholder, premium, and claims datasets, supporting regulatory reporting, risk assessment, and actuarial analysis for compliance
- Developed and maintained automated **ETL pipelines** using **Apache Spark**, **Hadoop**, and **Apache Airflow** to streamline ingestion, transformation, and loading of large-scale transactional and claims datasets, improving reporting efficiency.
- Designed and managed **data integration workflows** using **Apache NiFi** and **Talend** to consolidate policy administration, claims logs, and customer interaction data from multiple internal and third-party systems.
- Implemented scalable **data storage** and processing solutions with **MongoDB** and **Cassandra** to handle unstructured insurance documents, claim forms, and correspondence, ensuring secure and reliable data availability.
- Developed containerized analytics applications using **Docker** and managed **version control** with **Git**, enabling consistent deployment and reproducibility of predictive models and reporting solutions across development
- Leveraged **Snowflake** for centralized data warehousing, integrated with **Looker** for real-time dashboards, and utilized **Pandas**, **NumPy**, and **Matplotlib** for data cleaning, statistical analysis, and visualization of policy performance, claims trends, and customer insights.

Environment: Tableau, Power BI, Excel, Python, R, scikit-learn, TensorFlow, SQL, Apache Spark, Hadoop Apache Airflow, Apache NiFi, Talend, MongoDB, Cassandra, Docker, Git, Apache Hive, Snowflake, Looker, Pandas, NumPy, Matplotlib, Seaborn.